Introduction to LaTeX (Colours ,Figures and Picture Environment)

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Introducing Colour in LaTeX

LaTeX does not support coloured text by default. To use colour, load the xcolor package in your preamble.

Editor View:

LaTeX Code

```
\documentclass{article}
\usepackage{xcolor} % Enables colours in document
\begin{document}
Hello, \textcolor{red}{world}!
\end{document}
```

Document View:

Hello, world!

Basic Text Colours

Use \textcolor{color}{text} to colour entire words or phrases.
textbfEditor View:

LaTeX Code

```
\textcolor{blue}{Blue text} and
\textcolor{green}{green text}.
```

Document View:

Blue text and green text.

Colouring Mathematical Expressions

You can colour math symbols, expressions, or entire equations. **Editor View:**

LaTeX Code

```
\[
\textcolor{magenta}{E=mc^2}
\]
\[
\textcolor{orange}{\int_0^1 x^2\,dx}
\]
```

Document View:

$$\textcolor{magenta}{E=mc^2}$$

$$\textcolor{orange}{\int_0^1 x^2\,dx}$$

List of Predefined Colours

xcolor includes dozens of built-in colour names.

Editor View:

LaTeX Code

```
% Some built-in colours
\textcolor{red}{Red}, \textcolor{violet}{Violet},
\textcolor{teal}{Teal}, \textcolor{brown}{Brown},
\textcolor{cyan}{Cyan}, \textcolor{lime}{Lime}
```

Document View:

Red, Violet, Teal, Brown, Cyan, Lime

Defining Custom Colours

You can specify your own colours by RGB, HTML hex, or even by mixing existing colours.
textbfEditor View:

LaTeX Code

```
\definecolor{myblue}{RGB}{10,60,200}
\definecolor{softpink}{HTML}{FFB6C1}
\definecolor{alert}{named}{red!80!black}

\textcolor{myblue}{Custom Blue},
\textcolor{softpink}{Soft Pink},
\textcolor{alert}{Alert Red}
```

Document View:

Custom Blue, Soft Pink, Alert Red

Backgrounds and Highlights

Highlight text with background colour using `\colorbox{}` and outline text using `\fcolorbox{}`.

Editor View:

LaTeX Code

```
\colorbox{yellow}{Important highlight!} \\
\fcolorbox{red}{white}{Critical label}
```

Document View:

Important highlight!

Critical label

Changing Colour Mid-text and Scope

Use `{\color{color} ... }` to change colour for just a portion, or set the colour for the full document.

Editor View:

LaTeX Code

```
% Only this word is blue
{ \color{blue} BlueWord } is blue, while the
rest isn't.
% Make all subsequent text gray
\color{gray}
All the remaining text is gray from this point on.
```

Document View:

BlueWord is blue, while the rest isn't.

All the remaining text is gray from this point on.

Figures in LaTeX

Load graphicx and use `\includegraphics{filename}` to display images. The image file should be in your project.

Editor View:

LaTeX Code

```
\documentclass{article}
\usepackage{graphicx}
\begin{document}
\includegraphics[scale=0.5]{example2.jpg}
\end{document}
```

Document View:



To adjust the size of an image in LaTeX using , you can use the width, height, or scale options.

Placing Images with figure Environment and Captions

Use the figure environment for captions, centering, and cross-referencing.
Editor View:

LaTeX Code

```
\begin{figure}
   \centering
   \includegraphics[width=0.5\textwidth]
                           {example-image}
   \caption{A centered sample figure}
   \label{fig:sample}
\end{figure}
```

Document View:



Figure: A centered sample figure

Rotating and Cropping Images

Use angle=, trim=, and clip to rotate or crop. **Editor View:**

LaTeX Code

```
\includegraphics[angle=30]{example-image}
\includegraphics[trim=20 30 40 50, clip, width=3cm]{
```

Document View:





Multiple Images Side by Side

Use minipage or the subcaption package for layouts. **Editor View:**

LaTeX Code

```
\begin{minipage}{0.45\textwidth}
  \includegraphics[width=\linewidth]{exampleA}
  \captionof{figure:}{Image A}
\end{minipage}
\quad
\begin{minipage}{0.45\textwidth}
  \includegraphics[width=\linewidth]{exampleB}
  \captionof{figure:}{Image B}
\end{minipage}
```

Document View:



figure:Image A

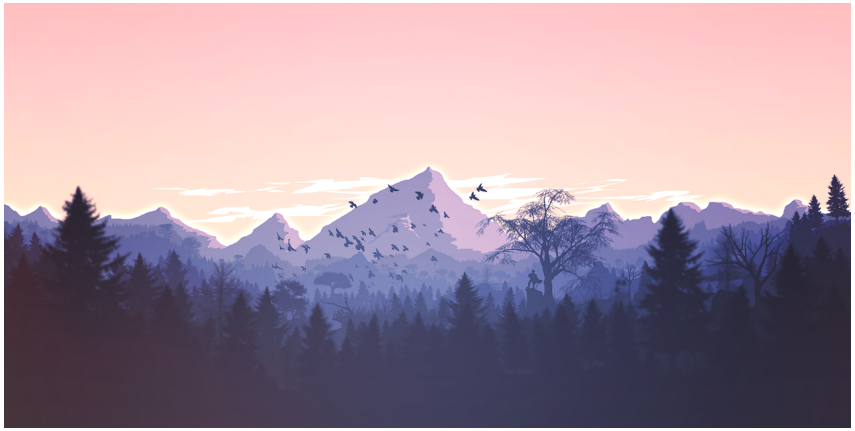


figure:Image B

Referencing Figures

Use `\label{...}` in your figure and reference later with `\ref{...}`.

Editor View:

LaTeX Code

```
See Figure ~\ref{fig:sample} for the result.
```

Document View:

See Figure 1 for the result.

The picture Environment — Drawing Coordinates

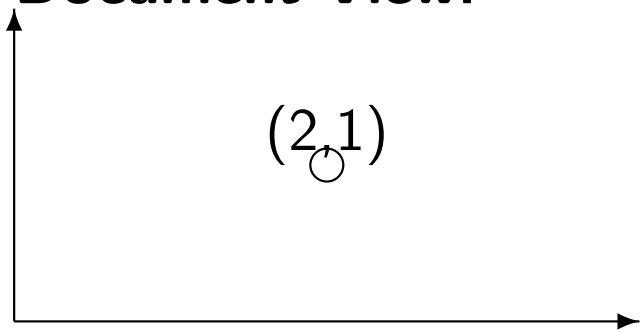
LaTeX's picture environment lets you draw objects (`lines, arrows, circles, etc.`) by specifying coordinates (x,y), measured in units set by `\unitlength`.

Editor View:

LaTeX Code

```
\setlength{\unitlength}{1cm}
\begin{picture}(4,2)
  \put(0,0){\vector(1,0){4}} % x-axis
  \put(0,0){\vector(0,1){2}} % y-axis
  \put(2,1){\circle{0.2}}
  \put(2,1){\makebox(0,0)[b]{(2,1)}}
\end{picture}
```

Document View:



A set of axes with an open circle marked at (2,1) and labeled.

Drawing Lines and Arrows

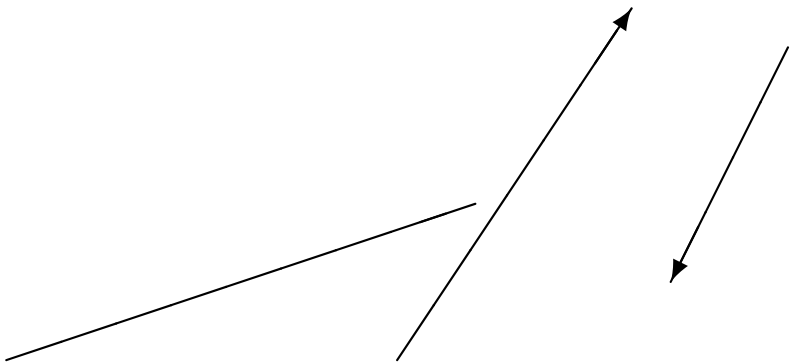
Use `\line()` and `\vector()` to draw straight lines or arrows. The pair (x,y) defines the slope, and the third argument gives horizontal length (except if vertical).

Editor View:

LaTeX Code

```
\setlength{\unitlength}{0.5cm}
\begin{picture}(8,6)
 \put(1,1){\line(3,1){6}} % a diagonal line
  \put(6,1){\vector(2,3){3}} % a  upward-pointing arrow
  \put(11,5){\vector(-1,-2){1.5}} % arrow going downward
\end{picture}
```

Document View:



Adding Shapes — Circles, Ovals, Frames

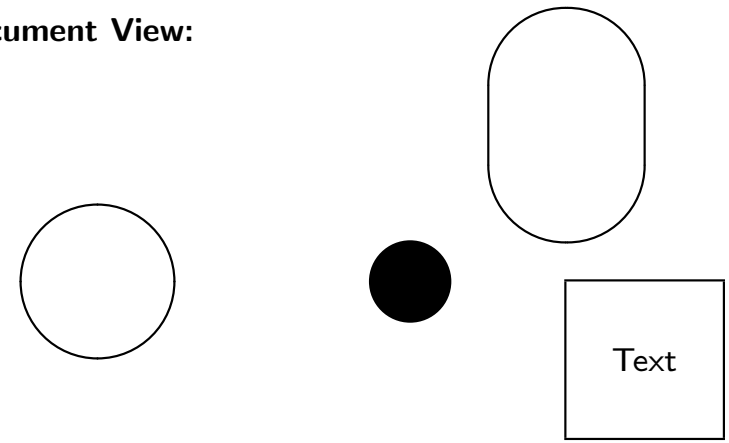
Draw circles, filled circles (disks), ovals, and insert framed boxes. Circles are positioned by their center; ovals by their center and width/height.

Editor View:

LaTeX Code

```
\setlength{\unitlength}{1cm}
\begin{picture}(5,3)
  \put(1,1.5){\circle{1}}
  \put(3,1.5){\circle*{0.6}}
  \put(4,2.5){\oval(1,1.5)}
  \put(4,0.5){\framebox(1,1)[c]{Text}}
\end{picture}
```

Document View:



Using Boxes and Positioning Text

You can place text or framed/makeboxes at specified coordinates with reference-point options (left, right, center, top, bottom).

Editor View:

LaTeX Code

```
\setlength{\unitlength}{1cm}
\begin{picture}(6,2)
  \put(1,1){\makebox(0,0)[l]{Left}}
  \put(3,1){\makebox(0,0)[r]{Right}}
  \put(5,1){\framebox(2,0.7)[t]{Top Frame}}
\end{picture}
```

Document View:



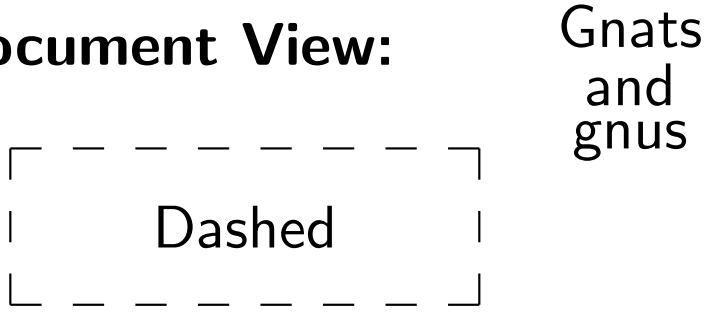
Dashed Boxes and Stacks

Editor View:

LaTeX Code

```
\setlength{\unitlength}{1cm}
\begin{picture}(5,2.5)
  \put(0.5,1){\dashbox{0.2}(3,1){Dashed}}
  \put(4,2){\shortstack{Gnats\\and\\gnus}}
\end{picture}
```

Document View:



Drawing Curves with

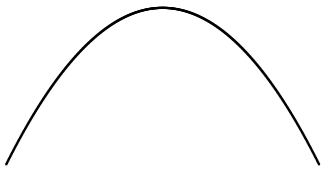
qbezier draws quadratic Bézier curves by specifying three control points.

Editor View:

LaTeX Code

```
\setlength{\unitlength}{0.1cm}
\begin{picture}(30,20)
  \qbezier(5,5)(15,25)(25,5)
\end{picture}
```

Document View:



Repeated Patterns

Use\multiput to repeatedly place an object with fixed displacements.

Editor View:

LaTeX Code

```
\setlength{\unitlength}{1cm}
\begin{picture}(6,2)
  \multiput(1,1)(1,0){5}{\circle*{0.2}}
\end{picture}
```

Document View:



Thank You!!!!!